

P1e: the continued-fraction induction — renormalisation of the
amplitude,
the triangle-loss obstruction, a hazard induction that proves $Z_N \neq 0$
on $[\frac{1}{5}, \frac{4}{5}]$,
and the seed 1/11
research programme P1, fifth atom, 2026-09-24

Notation of P1d_lemma.tex is used throughout. Cross-references use the numbering of the compiled P1d_lemma.tex, which has a single shared counter: Lemma 1 (W), Theorem 2 (Gauss factorisation $\Sigma_N = G_N^* e^{-\Phi(\varepsilon)} Z_N$), Lemma 4 (conj), Theorem 5 (Fresnel representation), Lemma 6 (dec), Definition 7 (main), Lemma 8 (lap), Theorem 9 (the corrected Lemma (1''- Δ)). (Earlier drafts of this note, and report_P1d.md, called Theorems 2, 5, 9 “Theorems 1, 3, 5”; corrected here, Reviewer AA.) Definition main (Main'_Z, S_{amp} , t_0 , F , L , B_k , γ_k). Labels: PROVED = complete proof here or in P1d; “computer-assisted” = reduced to finitely many Arb ball-arithmetic inequalities checked by the named script. VERIFIED = numerics, not a proof. HEURISTIC = argument without error control.

Outcome.

- (a) **Theorem M (PROVED, computer-assisted).** For every $N \geq 1$ and every a with $\gcd(a, N) = 1$ and $a/N \in [\frac{1}{5}, \frac{4}{5}]$ (equivalently $\ell \geq \log(4 \sin \frac{\pi}{5}) = 0.8548$): $Z_N(a/N) \neq 0$, hence $\Sigma_N \neq 0$ and $S_0 \neq 0$. Quantitatively $|Z_N| \geq 0.02 e^{-0.005N}$. The proof is an induction on N , not along the continued fraction but along a *hazard* decomposition, which is the continued-fraction recursion made N -free (Section 3).
- (b) **Item (ii) of the brief is exactly the induction hypothesis** (Lemma 2, PROVED): $S_{\text{amp}}(p/q; x) \rightarrow (1 - w_q)^{-1/(2q)} Z_q(p/q)$, with an explicit error bound. VERIFIED to 10^{-8} – 10^{-10} at 8 seeds.
- (c) **The obstruction to items (i),(iii) as posed** (VERIFIED): the error bound of Lemma lap sums errors over k with the triangle inequality. The loss $\Lambda = \sum_k |\gamma_k| |e^{B_k}| / |S_{\text{amp}}|$ grows exponentially along drift families ($\Lambda \approx 2.7 \cdot 10^7$ at 182/2001). Structured quantities stay bounded: $D_1 \leq 0.05$ and $D_2 \leq 0.003$ on the same families. A q -uniform version of Lemma lap therefore cannot work near drifted seeds. Theorem M avoids the loss by applying the seed lemma only in exponentially thin neighbourhoods, where the loss is beaten by the size of d . This works while $|u|e^{2\pi\eta} < 0.618$ (golden threshold), and the region $\ell < 0.48$ stays OPEN.
- (d) **Seed 1/11 fixed** (PROVED, computer-assisted). A sharpened outer-contour bound (Lemma 7) certifies 1/11 and 10/11 with $r = 10^{-5}$, $B \leq 0.315$ and 0.199 respectively, $n_2 = 600$ and $N^* \geq 3.1 \cdot 10^{13}$.

1 Exact finite formula and renormalisation

Lemma 1 (Zk; PROVED). *Let $x_0 = p/q$ in lowest terms, with $\ell(x_0) > 0$ and $4 \nmid q$ or $q\ell > \log 4$. At level q (i.e. $u = u_q = \lambda_q^2/c_0$, Φ_q excluding $q \mid n$),*

$$Z_q(p/q) = \sum_{n \geq 0} \psi_n \zeta_0^{-n^2} = \sum_{k \bmod q} \gamma_k e^{\Phi_q(\zeta_0^k)}, \quad \gamma_k = \frac{1}{q} \sum_{\nu \bmod q} \zeta_0^{-\nu^2 - k\nu}.$$

Proof. $n \mapsto \zeta_0^{-n^2}$ is q -periodic, so $\zeta_0^{-n^2} = \sum_k \gamma_k \zeta_0^{kn}$ (finite Fourier inversion). $\sum_n |\psi_n| < \infty$ because e^{Φ_q} is analytic on $|Y| < 1/|u_q|$ and $|u_q| < 1$. Exchange the sums. $\square$

Lemma 2 (R: the amplitude is the level- q object; PROVED). *In the setting of Definition main at the seed p/q , let $W_0 = Ue(-qt_0)$ and $X_k := -\frac{1}{2q} \log(1 - w_q) + \Phi_q(\zeta_0^k)$. Then*

$$\left| S_{\text{amp}} - (1 - w_q)^{-1/(2q)} Z_q(p/q) \right| \leq \left(\sum_k |\gamma_k| |e^{X_k}| \right) (e^{\delta_R} - 1),$$

with

$$\delta_R \leq \frac{|W_0| + |w_q|}{2q(1 - \max)} + \frac{q}{2} \epsilon_u \frac{y'}{1 - y'} + \frac{\pi}{4} q^2 d \frac{g}{1 - g} + \frac{q}{4} \frac{g^{n_2+1}}{(n_2 + 1)(1 - g)},$$

where $g \geq |u_0|, |u_q|$, $y' = (1 + \epsilon_u)g$, and $\epsilon_u = |\tilde{u}/u_q - 1|$ for $\tilde{u} = u_0 e(-t_0)$. Here $\epsilon_u \leq \exp(4\pi dg + |\lambda_N^2 - 1| + |\lambda_q^2 - 1| + 2\pi|t_0|e^{2\pi|t_0|}) - 1$. In particular $S_{\text{amp}} \rightarrow (1 - w_q)^{-1/(2q)} Z_q(p/q)$ as $d \rightarrow 0$, $N \rightarrow \infty$, $n_2 \rightarrow \infty$.

Proof. The saddle equation $\pi i t_0 + \frac{1}{q} \log(1 - W_0) = 0$ gives $e(-t_0) = (1 - W_0)^{2/q}$, so $\tilde{u}^q = U(1 - W_0)^2 = W_0$. At level q , $u_q^q = w_q = c_0^{-q}(1 - w_q)^2$. Hence $B_k(t_0) = -\frac{1}{2q} \log(1 - W_0) + \sum_{m \in A} \tilde{u}^m \zeta_0^{km} / (m(1 - \zeta^{-m}))$, and this differs from X_k in three ways:

- $\tilde{u}$ against u_q : $|\tilde{u}^m - u_q^m| \leq m\epsilon_u(1 + \epsilon_u)^{m-1}|u_q|^m$, with $|1 - \zeta^{-m}| \geq 2/q$ for $m \leq n_2$.
- ζ against ζ_0 in the denominators: $|\frac{1}{1 - \zeta^{-m}} - \frac{1}{1 - \zeta_0^{-m}}| \leq 2\pi m d \cdot \frac{q}{2} \cdot \frac{q}{4}$.
- The truncation $m > n_2$: $|1 - \zeta_0^{-m}| \geq 4/q$.

Then $|e^a - e^b| \leq |e^b|(e^{|a-b|} - 1)$, summed with weights $|\gamma_k|$, and Lemma 1 identifies $\sum_k \gamma_k e^{X_k}$. The bound on ϵ_u uses $|c_0/c - 1| \leq 4\pi d/|c|$, Lemma W at levels N and q , and $|e(-t_0) - 1| \leq 2\pi|t_0|e^{2\pi|t_0|}$. $\square$

VERIFIED (P1e_renorm.py, $N \approx 10^8$, $n_2 = 200$). Z_q by Lemma 1 agrees with $\Sigma_q/(G_q^* e^{-\Phi_q(\varepsilon)})$ to 10^{-25} – 10^{-51} . $|S_{\text{amp}}/((1 - w_q)^{-1/2q} Z_q) - 1|$ is: $1.5 \cdot 10^{-9}$ (1/3), $8 \cdot 10^{-10}$ (2/5), $8 \cdot 10^{-10}$ (3/7), $6 \cdot 10^{-8}$ (1/11), $9 \cdot 10^{-10}$ (4/9), $4 \cdot 10^{-10}$ (5/12), $3.5 \cdot 10^{-8}$ (2/21), $2 \cdot 10^{-9}$ (18/53). Every case is $O(d)$.

2 Why a q -uniform Lemma lap cannot close the induction (VERIFIED)

Lemma lap bounds $|Z_N/\text{Main}'_Z - 1|$ by $\sum_k |\gamma_k|(r_1 + \dots + r_5)/|S_{\text{amp}}|$. Each r_i carries a factor $\sup |e^{B_k}|$, so the bound carries the loss factor $\Lambda = \sum_k |\gamma_k| |e^{X_k}| / |Z_q|$. P1e_logderiv.py encloses (in Arb, rigorously) $|Z_q|$, Λ , and the moment ratios

$$D_1 = \frac{|f'(0)|}{2\pi q |f(0)|}, \quad D_2 = \frac{|f''(0)|}{(2\pi q)^2 |f(0)|}, \quad f(t) = \sum_k \gamma_k e^{\Phi_q(\zeta_0^k e(-t))}.$$

seed (family)	$ Z_q \geq$	$\Lambda \leq$	$D_1 \leq$	$D_2 \leq$
10/109 ($\theta = +1$ at 1/11)	1.057	17.9	0.047	0.0019
28/307	0.48	88	0.029	0.0023
91/1000	0.038	$1.0 \cdot 10^4$	0.035	0.0017
182/2001	$9.95 \cdot 10^{-4}$	$2.7 \cdot 10^7$	0.037	0.0016
182/2003 ($\theta = -1$ at 1/11)	$2.2 \cdot 10^3$	13.5	0.039	0.0014
191/2005 ($\theta = +1$ at 2/21)	0.79	97.5	0.0028	$5 \cdot 10^{-5}$
all $q \leq 150$, $\ell \geq 0.855$	0.91	13.6	0.11	0.031

Two points follow.

- Z_q itself drifts. The drift of Σ_N found in P1d is shared between $e^{-\Phi(\varepsilon)}$ and Z , and on the $\theta = +1$ side at 1/11 it is in Z . So Λ grows exponentially in q , like $e^{0.008q}$ here.
- D_1 and D_2 stay bounded.

Hence (HEURISTIC) the natural strengthened induction hypothesis for the full arc is a *moment* hypothesis: $|f^{(j)}(t)| \leq C_j(2\pi q)^j |f(0)|$ on $|t| \lesssim 1/q$. The error of the level- N main term would then be a structured sum, of relative size about $D_2(2\pi q)^2 d/A \approx 0.03 q/q'$, with no loss factor Λ . This route is not carried out here (OPEN). Theorem M below sidesteps it in the regime where it can.

3 The hazard induction

Fix $y = \frac{1}{2}$ and $\tau = \frac{1}{50}$. For a reduced fraction j/n define its *hazard radius*

$$\varepsilon_n = \begin{cases} \text{(low seeds, } n \leq 9) & \text{the certified radii of Table 1} \\ 2y^n/n^2 & 10 \leq n \leq 52 \\ y^n/(4\tau) & n \geq 53, \end{cases}$$

and the hazard $H(j/n) = \{x : 0 < |x - j/n| < \varepsilon_n\}$ (one-sided radii for the low seeds). Write the arc piece as the four regions $[\frac{1}{5}, \frac{1}{4}]$, $[\frac{1}{4}, \frac{1}{3}]$, $[\frac{1}{3}, \frac{2}{5}]$, $[\frac{2}{5}, \frac{1}{2}]$ (and mirror images). On region i put $g_i = \sup |u_0|$, which is 0.42535, 0.35357, 0.28869, 0.26288 respectively, and η_i with $g_i e^{2\pi\eta_i} \leq y$ ($\eta_i = 0.0257, 0.055, 0.087, 0.102$).

Lemma 3 (A: direct nonvanishing; PROVED). *Let $N \geq 151$ and $x = a/N$, and put $\mathcal{A} := \sum_{m \geq 1, N \nmid m} \frac{|u_0|^m}{m|1-\zeta^{-m}|}$. Then $|Z_N - 1| \leq e^{\mathcal{A}} - 1$. So $\mathcal{A} < \log 2$ implies $Z_N \neq 0$, with $|Z_N| \geq 2 - e^{\mathcal{A}}$.*

Proof. $Z_N = \sum_n \psi_n \zeta^{-n^2}$ with $\psi_0 = 1$. The series e^Φ is coefficientwise majorised by $\exp(\sum_m |a_m| Y^m)$, where $a_m = u_0^m/(m(1-\zeta^{-m}))$. Evaluate at $Y = 1$. $\square$

Lemma 4 (H: hazard combinatorics; PROVED). *Let $x = a/N \in [\frac{1}{5}, \frac{1}{2}]$.*

- (1) *Every hazard containing x has $n < N$, and the set of such hazards is finite.*
- (2) *Let j/n be the hazard of largest $n \geq 10$ containing x (deepest), $d = x - j/n$, and $n_2 = \lfloor 1/(2n|d|) \rfloor$. Then every $m > n_2$ that lies in the set T of Lemma dec satisfies $\|mx\| \geq m y^m/(4\tau)$, and so $\Delta_N(x; \rho) \leq \tau/n_2$ for every $\rho \leq y/g_i$.*
- (3) *If x lies in no hazard with $n \geq 10$ and in no low-seed disc, then $\|mx\| \geq m y^m/(4\tau)$ for all $m \geq 60$ with $N \nmid m$, and $\sum_{m \geq 60} |a_m| \leq \tau/59$.*
- (4) *If x lies in no hazard with $n \geq 10$ but in a low-seed disc of p/q (radius $r \leq 1/120$, $n_2 = 60$), then $\Delta_N(x; \rho_\eta) \leq \tau/60 < 10^{-3}$. So the Δ -hypotheses of Theorem 9 of P1d hold.*

Proof. (1) $|a/N - j/n| \geq 1/(nN)$ for $j/n \neq a/N$, and $\varepsilon_n < 1/(nN)$ once $n \geq N$ with $n \geq 10$.

(2),(3),(4) Suppose $\|mx\| < my^m/(4\tau)$. Then x is within $y^m/(4\tau)$ of some i'/m ; reduce it to i'/m' , which has $m' \leq m$. Three cases:

- $m' \geq 53$. Then $x \in H(i'/m')$, which contradicts deepest in (2) when $m' > n$, and contradicts the hypothesis in (3),(4).
- $10 \leq m' \leq 52$. Then $m \geq \max(60, 2m')$ (in (2) $m > n_2 \geq 2560$), and $y^{\max(60, 2m')}/(4\tau) \leq 2y^{m'}/m'^2$. So $x \in H(i'/m')$, with the same contradictions.
- $m' \leq 9$. In (3) this puts x in a low-seed disc. In (4), $i'/m' \neq p/q$ is impossible since $|p/q - i'/m'| \geq 1/81$; and $i'/m' = p/q$ forces $m = kq$ with $kqd < \frac{1}{2}$, i.e. $m \in M$, not T .

In (2) the reduced fraction i'/m' can also have $m' \leq n$, and the three cases above do not exclude this (a hazard with $m' \leq n$ containing x does not contradict “deepest”). Two sub-cases.

- $m' < n$ (shallower). Then $i'/m' \neq j/n$, so $|j/n - i'/m'| \geq 1/(nm')$ and $|x - i'/m'| \geq 1/(nm') - |d| \geq 1/(2nm')$, because $|d| < \varepsilon_n$ and $nm'\varepsilon_n \leq \frac{1}{2}$. That is incompatible with $|x - i'/m'| < y^m/(4\tau)$ for $m > n_2 \geq 2560$.
- $m' = n$. If $i' \neq j$, then $|x - i'/n| \geq 1/n - |d| \geq 1/(2n)$, again incompatible. If $i'/n = j/n$, then $|d| = |x - j/n| < y^m/(4\tau)$ with $m > n_2 = \lfloor 1/(2n|d|) \rfloor$, so $m \geq 1/(2n|d|)$ and $D := |d|$ satisfies $Dy^{-1/(2nD)} < \frac{1}{4\tau} = 12.5$. The function $f(D) = D2^{1/(2nD)}$ decreases on $(0, \log 2/(2n)]$, and $\varepsilon_n < \log 2/(2n)$, so $f(D) \geq f(\varepsilon_n)$ on $(0, \varepsilon_n)$. For $n \geq 53$: $f(\varepsilon_n) = 12.5 \cdot 2^{-n+2^n/(25n)} \geq 12.5$, since $2^n \geq 25n^2$. For $10 \leq n \leq 52$: $1/(2n\varepsilon_n) = n2^n/4$ and $f(\varepsilon_n) = 2n^{-2}2^{-n+n2^n/4} \geq 12.5$, since $n2^n/4 \geq n + 2\log_2 n + 3$. Either way $f(D) \geq 12.5$, a contradiction.

(The multiples $m = kn$ with $k \leq n_2$ are not in T : they form the set M of Lemma 6 (dec) of P1d at the seed j/n , whose $m_c = \lfloor 1/(2n|d|) \rfloor$ equals n_2 . The case $m' = n$, $i' = j$ is exactly the case of the remaining multiples $m = kn$, $k > n_2$, which the argument above excludes. This sub-case was missing from the earlier write-up; Reviewer AA.) So every m in question has $\|mx\| \geq my^m/(4\tau)$, and the term is at most $(|u_0|\rho)^m \tau / (m^2 y^m) \leq \tau / m^2$, since $|u_0|\rho \leq g_i \rho \leq y$. Sum over m . $\square$

Lemma 5 (B: uniform deep-seed estimate; PROVED, explicit). *Let $j/n \in \text{region } i$ with $n \geq 53$, and $0 < |d| \leq y^n/(4\tau)$. Assume $\Delta_N(x; e^{2\pi\eta_i}) \leq \tau/n_2$ and $|Z_n(j/n)| \geq c_0 e^{-\kappa n}$, where $c_0 = 0.02$ and $\kappa = 0.005$. Then*

$$|Z_N/\text{Main}'_Z - 1| \leq E(n) \leq \frac{1}{2}, \quad |S_{\text{amp}}| \geq (1 - R(n))(1 - w_n)^{-1/2n} |Z_n(j/n)|, \quad R(n) \leq \frac{1}{2},$$

where E, R are the explicit functions evaluated by `P1e_lemmaB.py`.

Proof. Apply Lemma 8 (lap) of P1d with $v_0 = \eta/2$, t_0 the unique zero of F' in $|t| \leq \eta$ (a contraction: $|t_0| \leq R_0 = \bar{w}/((1 - \bar{w})\pi n)$, $\bar{w} = (g\rho)^n \leq y^n$), and the following bounds valid on $\text{Im } t \leq \eta$.

- $|W| \leq \bar{w}$.
- $A \geq \frac{\pi}{2} \frac{1 - \bar{w}}{1 + \bar{w}}$ and $|\alpha - \pi/4| \leq \arcsin \frac{\bar{w}}{1 - \bar{w}}$.
- $K_3 \leq \frac{4\pi^2 n \bar{w}}{6(1 - \bar{w})^2}$ and $K_4 \leq \frac{8\pi^3 n^2 \bar{w}(1 + \bar{w})}{24(1 - \bar{w})^3}$.
- For $m \leq n_2$ with $n \nmid m$: $\|mx\| \geq \frac{1}{n} - m|d| \geq \frac{1}{2n}$, hence $|c_m e(-mt)| \leq ny^m/(2m)$. This gives $|e^{B_k}| \leq \beta = \exp(\frac{\bar{w}}{2n(1 - \bar{w})} + \frac{n}{2}L)$ with $L = -\log(1 - y)$, $|B'_k| \leq P_1 = \frac{\pi \bar{w}}{1 - \bar{w}} + \frac{\pi ny}{1 - y}$, and $|B''_k| \leq P_2 = \frac{2\pi^2 n \bar{w}}{(1 - \bar{w})^2} + \frac{2\pi^2 ny}{(1 - y)^2}$.

The Δ -terms are bounded by τ/n_2 . The gap uses $X_1 \geq -R_0 + (\eta - R_0) \cot \alpha_{\max}$ and $C_0 \leq 2 \frac{\bar{w} + \bar{w}^2/4(1-\bar{w})}{2\pi n^2} + \pi R_0^2$. $|S_{\text{amp}}|$ is bounded below by Lemma 2 with $\sum |\gamma_k| |e^{X_k}| \leq \sqrt{2n} \beta$, since $|\gamma_k| \leq \sqrt{2/n}$. In ϵ_u one uses $N \geq 1/(n|d|)$, which makes $|\lambda_N^2 - 1|$ doubly exponentially small. All bounds increase with $|d|$ and are evaluated at $|d| = y^n/(4\tau)$. Numerical evaluation: `P1e_lemmaB.py` evaluates every quantity in Arb and checks all side conditions ($K_3 v_0 \leq A/2$, $A' > 0$, $\text{gap} > 0$, $d \leq 2\text{gap}$, $n_2 \geq n$, $n\ell \geq 3$, $g_i e^{2\pi\eta_i} \leq y$) and $E, R \leq \frac{1}{2}$. The check covers all $53 \leq n \leq 3000$ and all four regions: 2948 values each, 0 failures; the maximum is $E(53) \leq 0.5$. Tail $n > 3000$: here $\bar{w} \leq 2^{-n}$, $\beta \leq 1.01 \cdot 2^{n/2}$, $d \leq 12.5 \cdot 2^{-n}$ and $n_2 \geq 2^n/(25n)$. Then $r_1 \leq 200n^2 2^{-n/2}$ and $r_3 \leq 2n 2^{-n/2}$, while $r_2, r_4, r_5 \leq \exp(-2^{n/2})$. Also $R \leq 10^4 n^{5/2} (e^\kappa/\sqrt{2})^n$ and $E \leq 10^6 n^{5/2} (e^\kappa/\sqrt{2})^n$, with $e^\kappa/\sqrt{2} = 0.7107$. So $E, R < 10^{-400}$ for $n > 3000$. Spot values are consistent with this: $E(10^4) \leq 1.7 \cdot 10^{-1471}$, $E(10^8) \leq 3 \cdot 10^{-14839060}$. $\square$

Lemma 6 (B': seeds $10 \leq n \leq 52$; PROVED, computer-assisted). *For every reduced $j/n \in [\frac{1}{5}, \frac{4}{5}]$ with $10 \leq n \leq 52$ (480 seeds), side $d > 0$ (side $d < 0$ is the seed $(n-j)/n$ by Lemma conj), and $0 < d \leq 2y^n/n^2$: the conclusion of Lemma 5 holds with $E \leq 0.0019$ and $R \leq \frac{1}{2}$. Here β, P_1, P_2 are replaced by the exact sums over $m \leq 200$ (with $|1 - \zeta^{-m}| \geq |1 - \zeta_0^{-m}| - 2\pi m d$), and $|Z_n(j/n)|$ and $\sum |\gamma_k| |e^{X_k}|$ are enclosed exactly by Lemma 1. Script: `P1e_lemmaBp.py`; all 480 seeds pass.*

Lemma 7 (O: sharpened outer contour; PROVED). *In Lemma lap the term r_4 may be replaced by $(r_{4a} + r_{4b} + r_{4c})\beta_\eta e^{\delta_\eta}$, and the horizontal term by r_5 with $X_1 \rightarrow X_2$ plus r_{5h} . Here:*

- $r_{4a} = \sqrt{A/A_{\text{loc}}} \operatorname{erfc}(v_0 \sqrt{A_{\text{loc}}/d})$ with $A_{\text{loc}} = A - K_3^{\text{loc}} V_q$. This rests on $\operatorname{Re}(F - F_0) \leq -Av^2 + K_3^{\text{loc}} |v|^3$ on $|v| \leq V_q$.
- r_{4b} and r_{5h} are the sums $\sum_j \text{len}_j e^{-\kappa_j/d} \sqrt{A/(\pi d)}$ over sub-balls of the upper ray and of $[X_1, X_2] + i\eta$, where $\kappa_j \leq -\operatorname{Re}(F - F_0)$ is Arb-enclosed and $d \leq 2\kappa_j$ is checked.
- r_{4c} is the same for the lower ray on $[-V_{\text{neg}}, -V_q]$, plus $\frac{1}{2} \sqrt{A/c_q} \operatorname{erfc}(V_{\text{neg}} \sqrt{c_q/d})$. This rests on $\operatorname{Re}(F - F_0) \leq -\frac{\pi}{2} \sin 2\alpha v^2 + \pi |t_0| |v| + 2L_{\text{max}}$ on the lower ray, where $|W| \leq |W_0|$.

Proof. Each piece bounds $\int |e^{(F-F_0)/d}| |dt|$ on its part of the same contour Γ as Lemma lap, and together the pieces cover Γ . The ray pieces use $|e^{B_k}| \leq \beta_\eta$ and $|e^\epsilon| \leq e^{\delta_\eta}$ exactly as r_4 does. $\square$

Implemented in `P1e_cert.py`, which is `P1d_cert.py` with Lemma 7.

(Directed rounding, Reviewer AA: every entry is printed by `dround.py` from the Arb ball, lower bounds ($|S_{\text{amp}}|$, c_{seed}) rounded *down* and upper bounds (B , κ_{seed}) rounded *up* in the last printed digit; c_{seed} is computed in Arb from the rounded-down $|S_{\text{amp}}|$. Raw output: `lowseed_directed.txt`. The smallest c_{seed} is 0.1921 (seed 3/4), a factor 9.6 above the needed $c_0 = 0.02$; the largest κ_{seed} is $0.00132 < \kappa = 0.005$.)

Theorem 8 (M; PROVED, computer-assisted). *For all $N \geq 1$ and $\gcd(a, N) = 1$ with $a/N \in [\frac{1}{5}, \frac{4}{5}]$: $|Z_N(a/N)| \geq c_0 e^{-\kappa N}$, where $c_0 = 0.02$ and $\kappa = 0.005$. In particular $\Sigma_N \neq 0$ and $S_0 \neq 0$ there.*

Proof. By Lemma conj it suffices to take $x = a/N \in [\frac{1}{5}, \frac{1}{2}]$. Strong induction on N .

Base $N \leq 150$. `P1e_logderiv.py scan 2 150 arc` (run in the four blocks 2–70, 71–105, 106–130, 131–150) encloses $Z_N(a/N)$ by Lemma 1 for every reduced a/N with $N \leq 150$ and $N \leq 5a \leq 4N$ (an exact integer test), 4117 cases, giving $|Z_N| \geq 0.9134$ (rounded down). The scan fails loudly: an exception or a lower bound for $|Z_N|$ that is not certified positive aborts it with exit status 1 and the message `SCAN FAILED`; each block ended with `SCAN PASSED` and exit status 0 (`base_arc*.txt`). This was tested by injecting a failure at 2/5. (Nonvanishing of Σ_N for $N \leq 150$ is also P1b's interval certificate.)

Step $N \geq 151$. Four cases.

Table 1: Low seeds (`P1e_lowseed.py`; side $d > 0$; $n_2 = 60$; the $d < 0$ side of p/q is the row $(q - p)/q$). $c_{\text{seed}} = (1 - B)\sqrt{\pi/2A}|S_{\text{amp}}|$ and $\kappa_{\text{seed}} = q \max(0, -\text{Re } F(t_0))$.

seed	r	η	$B \leq$	$\kappa_{\text{seed}} \leq$	$ S_{\text{amp}} \geq$	$c_{\text{seed}} \geq$
1/2	0.0083	0.102	0.4945	0.000270	1.142	0.5463
1/3	0.004	0.087	0.2927	0.00132	1.087	0.7682
2/3	0.003	0.055	0.6473	0	1.092	0.3847
1/4	0.003	0.055	0.7730	9.79e-6	1.043	0.2406
3/4	0.0007	0.0257	0.8204	2.10e-6	1.052	0.1921
1/5	0.0005	0.0257	0.4907	0	0.9679	0.4927
2/5	0.003	0.102	0.05722	4.10e-5	1.084	1.022
3/5	0.003	0.087	0.06302	0	1.086	1.018
2/7	0.002	0.055	0.3019	0	1.075	0.7511
5/7	0.002	0.055	0.3032	8.22e-6	1.077	0.7511
3/7	0.002	0.102	0.02728	1.69e-6	1.073	1.044
4/7	0.002	0.102	0.02695	0	1.074	1.046
3/8	0.002	0.087	0.02049	9.70e-9	1.094	1.072
5/8	0.002	0.087	0.01971	1.02e-8	1.095	1.074
2/9	0.0005	0.0257	0.5569	3.63e-6	1.018	0.4515
7/9	0.0005	0.0257	0.5579	0	1.018	0.4505
4/9	0.0015	0.102	0.02223	7.86e-8	1.066	1.043
5/9	0.002	0.102	0.03172	0	1.067	1.034

1. x lies in a hazard with $n \geq 53$. Take the deepest hazard j/n . Lemma 4(2) gives the Δ -hypothesis. The induction hypothesis at level $n < N$ gives $|Z_n(j/n)| \geq c_0 e^{-\kappa n}$. Lemma 5 then gives $|Z_N| \geq \frac{1}{2} |\text{Main}'_Z| \geq \frac{1}{4} \sqrt{\pi/2A} c_0 e^{-\kappa n} e^{-n|F(t_0)|N}$. Here $N \geq 1/(n|d|) \geq 0.08 \cdot 2^n/n$ and $n|F(t_0)| \leq 2^{-n}$, so this is at least $c_0 e^{-\kappa N}$.
2. The deepest hazard has $10 \leq n \leq 52$. The same argument, with Lemma 6 and the exact $|Z_n| \geq 0.91$.
3. x lies in no hazard with $n \geq 10$ but in a low-seed disc. Lemma 4(4) and Theorem 9 of P1d (the certificate of Table 1) give $|Z_N| \geq c_{\text{seed}} e^{-\kappa_{\text{seed}} N} \geq c_0 e^{-\kappa N}$.
4. Otherwise. `P1e_caseA.py` covers $[\frac{1}{5}, \frac{1}{2}]$ minus all hazards of height ≤ 59 and the low-seed discs, by 22,546 Arb intervals with 0 failures. It shows $\sum_{m < 60} \sup |a_m| + \tau/59 \leq 0.6057$. (Correction, Reviewer AC: the script had excluded the hazards of height $53 \leq n \leq 59$ with radius $25 \cdot 2^{-n}$, twice $\varepsilon_n = y^n/(4\tau)$, leaving the annuli $\varepsilon_n \leq |x - j/n| < 2\varepsilon_n$ uncovered. Fixed and rerun 2026-09-24: still 22,546 intervals, 0 failures, bound 0.6057.) With Lemma 4(3) and Lemma 3 this gives $|Z_N| \geq 2 - e^{0.6057} = 0.167$.

The cases are exhaustive. Theorem 2 of P1d converts the result to Σ_N , and S0_lemma converts it to S_0 . $\square$

Remark 9 (scope and the golden threshold). The only non-uniform input of Lemma 5 is the crude bound $\sup |e^{B_k}| \leq e^{nL/2}$. Against the hazard size y^n it closes iff $y e^{L/2+\kappa} < 1$, i.e. $y < (\sqrt{5} - 1)/2$ as $\kappa \rightarrow 0$. The method therefore extends, with more low seeds, up to about $4 \sin \pi x > 1.618$, i.e. $\ell > 0.48$. Below that, the loss Λ of Section 2 must be removed. That needs the structured (moment) induction hypothesis; this remains OPEN, and so does the edge $\ell < 0.48$, including the $1/11$ region ($\ell = 0.12$).

Remark 10 (direct argument). Lemma 3 is the direct zero-free argument ($Z = 1 + \text{small}$). It covers all but 2.6% of $[0.2, 0.5]$ (VERIFIED on a grid), and 0% of $\ell < 0.21$, because there $|u|^1$ alone exceeds the

budget. An argument-principle proof from Theorem 5 of P1d was not found; the integrand e^Φ has no positivity. (HEURISTIC assessment.)

Seed 1/11 (PROVED, computer-assisted). `P1e_cert.py 1 11 1e-5 600 0.012 0.008 4 6 1e-3 1 32 0.4 48` gives $B \leq 0.3150$ and $N^* \geq 3.36 \cdot 10^{13}$. The same arguments with `10 11` give $B \leq 0.1994$ and $N^* \geq 3.12 \cdot 10^{13}$. P1d failed at this seed with $A' \leq 0$, because the single global quadratic bound wastes $|L''(t_0)| \approx A$ when $|U| = 0.27$. Lemma 7 replaces it by a cubic local bound and Arb-enclosed pieces. VERIFIED end to end: `P1d_sanity.py 1 11 600 10000` gives $|\Sigma_N/\text{Main}'_N - 1| = 6.05 \cdot 10^{-4}$ at $|d| = 9.1 \cdot 10^{-6}$, so $K(1/11) \approx 67$, against a certified 0.315.

Files. `P1e_renorm.py` (Lemma R), `P1e_logderiv.py` (Lemma Zk enclosures; Λ, D_1, D_2 ; base scan), `P1e_cert.py`, `P1e_all.py` (Lemma O certificate and search), `P1e_lowseed.py` (Table 1), `P1e_lemmaB.py` (Lemma B), `P1e_lemmaBp.py` (Lemma B'), `P1e_caseA.py` (case A covering), `P1e_certlog.txt` (raw outputs), `dround.py` (directed-rounding printing), `lowseed_directed.txt`, `base_arc_*.txt` (reruns with directed rounding and the fail-loud scan, P1f). Every run used `perl -e 'alarm 290; exec @ARGV'` and stayed below about 300 MB.